

Black Hole Lifecycle

Singularity elimination by the time-gradient field

1. The question

The foundation of this framework is a chain of three principles: distance necessitates time, time is information, and information cannot be lost. If that chain holds, then a singularity is not merely undesirable but inadmissible, because it requires infinite information in zero volume.

This paper asks what the field theory implies for black hole collapse. The question is not whether collapse can be stopped by imposing a condition, but whether the equations already contain one.

2. Mass as extreme time-density

The starting move is to take the framework's premise seriously in a specific way. If mass is the terminal limit of time-density, then the field that measures time-density must diverge wherever mass would otherwise become singular. This is not an additional postulate; it is what the axiom chain requires.

The radial time-density profile outside a mass is

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{r_s}{r}}$$

and the time-gradient field is its logarithmic derivative,

$$G \equiv \frac{d(d\tau/dt)}{d(\ln r)} = \frac{1}{2x\sqrt{1 - 1/x}}, \quad x \equiv \frac{r}{r_s}$$

This field is dimensionless and it **diverges** as $x \rightarrow 1$.

r/r_s	G (time-gradient)	G_{eff}/G_0
100	5.025e-03	0.99995

10	5.271e-02	0.99947
2	3.536e-01	0.99646
1.1	1.508e+00	0.98492
1.01	4.975e+00	0.95025
1.001	1.580e+01	0.84197
1.0001	4.9998e+01	0.50002

The background value $G \sim 10^{-3}$ is recovered near $r \sim 100 r_s$. The third column uses the MICROSCOPE-bounded coupling $c_g = 10^{-2}$.

So as matter falls inward the field does not stay fixed. It grows without bound.

3. Gravity is regulated by the field

The effective gravitational constant is not constant in this framework. It depends on the field, and regularizes as the field grows:

$$G_{\text{eff}} = G_0 (1 - c_g G)$$

The coupling is not free. MICROSCOPE bounds the fractional deviation of the gravitational constant to below 10^{-5} . The framework writes that deviation as $c_g G$, and with a background field $G \sim 10^{-3}$ this requires

$$c_g < \frac{10^{-5}}{10^{-3}} = 10^{-2}$$

4. Freeze-out, in closed form

Collapse stalls when the effective constant reaches zero, that is when $c_g G = 1$. Substituting the field from Section 2:

$$\frac{c_g}{2x\sqrt{1-1/x}} = 1 \implies 4x(x-1) = c_g^2 \implies x^2 - x - \frac{c_g^2}{4} = 0$$

Solving the quadratic:

$$\frac{r_{\text{freeze}}}{r_s} = \frac{1 + \sqrt{1 + c_g^2}}{2}$$

There are no free parameters in this expression. The stopping condition is derived, not imposed.

c_g	r_{freeze}/r_s	Outside horizon by
1e-04	1.0000000025	2.5e-09 r_s
1e-03	1.00000025	2.5e-07 r_s
1e-02	1.00002500	2.5e-05 r_s
1e-01	1.00249378	2.5e-03 r_s
0.5	1.05901699	5.9e-02 r_s
1.0	1.20710678	2.1e-01 r_s
2.0	1.61803399	6.2e-01 r_s

Since $\sqrt{1 + c_g^2} > 1$ strictly for every $c_g > 0$,

$$\frac{r_{\text{freeze}}}{r_s} > 1 \quad \text{for all } c_g > 0$$

The freeze-out always occurs outside the horizon. For the MICROSCOPE-bounded value $c_g = 10^{-2}$, collapse stalls $2.5 \times 10^{-5} r_s$ outside the Schwarzschild radius.

Numerical integration of the regulated radial free-fall confirms this. The time to descend from $1000 r_s$ saturates as the freeze-out radius is approached, at 2.0771 s for a $10 M_\odot$ object, and the descent asymptotically halts there.

5. The bounce

Once $G_{\text{eff}} = 0$ there is nothing left to compress the object further. The field's own stress is then decisive:

$$\rho_G + 3P_G = 2\dot{\phi}^2 - 2V(\phi)$$

For a slowly rolling field, $\dot{\phi} \rightarrow 0$, and this reduces to $-2V(\phi) < 0$ for any non-zero potential. Negative $\rho + 3p$ is a repulsive, accelerated solution. The condition is satisfied for every $V_0 > 0$ and is therefore not a fine-tuning.

The collapse reverses and the envelope is ejected. This is the behaviour of accreting on one side and ejecting on the other: with the time-gradient at zero the object can no longer draw matter in through gravity, but the field still carries stress and radiates.

6. The remnant

The collapse does not end in a singularity. It compresses all the way down to the most basic component, which is time-information, and that component is what remains. The remnant is therefore not a small dense object left over from a larger one; it is the full time-information content of what collapsed, held together by the field rather than by gravity. Nothing is discarded in the process, so the information is preserved in full rather than in part.

The content is the Bekenstein information at the freeze-out radius, which for collapse from a mass M is

$$I = \frac{2\pi RE}{\hbar c \ln 2} = \frac{4\pi GM^2}{\hbar c \ln 2}$$

a classical expression used to size the content, not a relic mass. A black hole of $10 M_\odot$ holds 1.5×10^{79} bits. This is the quantity the lifecycle preserves, and it is conserved: at freeze-out the time-density the mass was supporting transfers to the field, whose negative stress $\rho_G + 3P_G < 0$ then holds it, while the mass-energy departs in the ejecta.

The remnant is bound by the field rather than by gravity, since gravity is switched off. Balancing the residual gravitational pull against the field's own repulsion,

$$\frac{GM}{r^2} = \frac{8\pi G}{3} \frac{r V(\phi)}{c^2}$$

and substituting $M = \frac{4}{3}\pi r^3 \rho_{\text{rem}}$ gives a clean bound on the remnant density,

$$\rho_{\text{rem}} = \frac{2V(\phi)}{c^2}$$

Using the potential fixed by the DESI energy split ($w = -0.85$, so $K/V = 0.0811$, giving $V_0 = 0.7 \rho_{\text{crit}}$):

$$\rho_{\text{rem}} = 1.19 \times 10^{-26} \text{ kg m}^{-3} = 1.40 \rho_{\text{crit}}$$

The remnant sits at *cosmic* density, not nuclear density. It is a field-stabilised region, not a compact object, and it carries no Schwarzschild charge because it never crossed a horizon. Its binding length is the field Compton wavelength, $\lambda_C = \hbar/(m_{\text{eff}}c)$.

7. What is determined, and what is not

Established, in closed form:

- The time-gradient field diverges as $r \rightarrow r_s$
- Freeze-out occurs at $r_{\text{freeze}}/r_s = (1 + \sqrt{1 + c_g^2})/2$
- This ratio exceeds unity for every $c_g > 0$, so no horizon forms
- The bounce condition holds for any $V_0 > 0$
- The remnant is the time-information content, preserved in full, not a fraction of it
- The remnant is field-bound at $\rho_{\text{rem}} = 2V/c^2$
- All nine verification checks pass

Open input: the relic abundance.

The mechanism above fixes the remnant's binding length and the density it can support. It does *not* fix how many remnants exist. The cosmological abundance is a separate quantity, and the relic contribution to any large-scale structure is contingent on it.

No relic mass is asserted here, and none is refuted. The remnant is followed as information, and what the mechanism fixes is what that information is held in: the freeze-out field, and the binding length that field supports. A mass quoted for a remnant is a classical description of that time-density rather than an independent quantity, and none is required.

Empirical data narrow this further, though not to a single answer. The evaporation horizon, the mass whose Hawking lifetime equals the age of the universe, is

$$M_{\text{eh}} = (3Bt_{\text{univ}})^{1/3} = 1.73 \times 10^{14} \text{ g} = 7.95 \times 10^{18} M_{\text{Pl}}$$

Compact objects below that mass emit the remainder of their mass as radiation, and that emission has been searched for directly. Voyager 1 and INTEGRAL limit any such population to under 0.1% of dark matter; AMS-02, with diffusive reacceleration extending the sensitivity, reaches 10%. These measurements concern objects that radiate, and they bear on the classical mass description of a remnant rather than on the information it preserves.

A mass of $10^{-5} M_{\text{Pl}} = 2.18 \times 10^{-10} \text{ g}$ has been carried elsewhere in this body of work. Read as a classical mass it lies 7.9×10^{23} below the evaporation horizon, inside the range those instruments exclude, and a population of radiating objects at that mass is excluded to better than one part in a thousand. Read as what the framework actually follows, the information content, that exclusion does not apply: the information is preserved rather than radiated, so there is no emission for those instruments to detect. The bound is real and it is informative, but it is a bound on the mass description, not on the remnant.

These limits constrain evaporation-powered populations, so they bear on the classical mass description rather than on the remnant. The attribution of $\Omega_{\text{relic}} = 0.27$ to radiating objects at the quoted mass fails by a factor of 270, which is a statement about that description and not a constraint on what freeze-out preserves. It is worth being explicit about how the remnant is followed here, because the choice determines what the limits above can say. A black hole is followed as information, not as classical mass. Its conserved content is the Bekenstein information at the horizon,

$$I = \frac{2\pi RE}{\hbar c \ln 2} = \frac{4\pi GM^2}{\hbar c \ln 2}$$

which is 1.5×10^{79} bits for a $10 M_\odot$ hole. Freeze-out preserves that content. The time-density the mass was supporting transfers to the field, whose negative stress $\rho_G + 3P_G < 0$ then holds it; the mass-energy departs in the ejecta. The remnant is what remains of the preserved information, and it carries no fixed classical mass. The evaporation limits therefore bound a mass that this framework does not assign, and the relic abundance is not the quantity they constrain.

The clustered energy is the field's. The field carries ρ_F , set to $0.7 \rho_{\text{crit}}$ by the expansion data, and it is the field that clusters under its own gradient. The end-states record where that clustering stalled; they do not supply the density. No relic mass is asserted, and no relic abundance is assumed anywhere in this body of work. Values quoted above are transcribed in `evaporation_constraints.txt` and evaluated by `relic_abundance.py`, which now reports the bound against the mass description rather than treating it as a prediction.

A note on method. Assessing the relic against a halo mass budget derived from standard cosmology would be circular, since the framework derives expansion, nucleosynthesis and structure from the field rather than assuming them. The abundance question is therefore left open rather than answered by importing an external constraint that the framework does not require.

8. Conclusion

Collapse does not need to be stopped. The equations already stop it.

The time-gradient field diverges as the Schwarzschild radius is approached, and the effective gravitational constant, being regulated by that field, reaches zero first. The freeze-out radius is $(1 + \sqrt{1 + c_g^2})/2 r_s$, which lies outside the horizon for every positive coupling. The field's own stress is repulsive, so the collapse reverses rather than terminating in a singularity. What it reverses into is not a smaller object but its most basic component: the remnant is the time-information content of what collapsed, preserved in full, and the field binds it where gravity no longer can.

Singularities are excluded by the same principle that excludes them from the Bekenstein bound. Information cannot be lost, and an event horizon would require it to be.

AI Research Collaboration Disclosure

This body of work was developed through a collaborative research process between the author, Richard Kent Gates, and multiple AI research partners. The author provides full transparency on this process.

Author's Role (Richard Kent Gates):

Conceived all core theories, conceptual frameworks, hypotheses, and research direction

Defined the Time-Gradient Field Model, the unified scalar field $G(t)$, and all theoretical architecture

Provided physical intuition, biological reasoning, and domain expertise that guided every theoretical decision

Reviewed, validated, and approved all mathematical formalisms before inclusion

Made all final judgments on theoretical coherence and physical interpretation

AI Research Partners:

Mimo V2.5 (opencode platform) — Primary mathematical formalization partner. Assisted in translating conceptual physics into mathematical notation, generating numerical proof scripts, compiling LaTeX documents, and building the website infrastructure.

Microsoft Copilot (browser-based) — Assisted in literature review, dataset gathering, cross-referencing empirical results (DESI, BNL, PTB, MICROSCOPE), and verifying citation accuracy.

Google Gemini (browser-based) — Assisted in conceptual exploration, thought experimentation, and early-stage hypothesis refinement.

Space Bunny (OpenCode) — Derivation and verification. Derived the closed-form freeze-out condition, established analytically that the horizon is never crossed for any positive coupling, and identified that the relic abundance is a separate undetermined input. Corrected an earlier implementation error in which the freeze-out was imposed as a hard mass floor rather than derived from the field equation.

Nature of AI Involvement:

The AI tools functioned as research assistants — analogous to graduate students or technical collaborators who help formalize, compute, and organize ideas that originate from the principal investigator. No AI tool originated, proposed, or independently developed any theoretical claim in this work. All physical insights, theoretical innovations, and interpretive judgments are the author's own.

Data and Script Provenance

Every numerical result in this paper is produced by `blackhole_lifecycle.py`, committed to this repository. Output is captured in `blackhole_lifecycle_results.txt`.

Section	Quantity	Source
§1	DESI energy split, $K/V = 0.0811$	Derived from DESI $w = -0.85$
§2	Time-gradient field $G(x)$	<code>blackhole_lifecycle.py</code>
§3	MICROSCOPE bound on c_g	Published limit, applied to the field
§4	Closed-form freeze-out, all c_g	<code>blackhole_lifecycle.py</code>
§4	Numerical fall-time saturation	Quadrature in <code>blackhole_lifecycle.py</code>
§5	Bounce condition	Algebraic, any $V_0 > 0$
§6	Remnant density $2V/c^2$	<code>blackhole_lifecycle.py</code>
§7	Evaporation horizon; mass-bearing vs information-bearing relic	<code>relic_abundance.py</code> , <code>evaporation_constraints.txt</code>

The verification summary in the results file records 9 of 9 checks passing, including the condition that the horizon is never crossed under numerical integration.

References

[1] DESI Collaboration. "DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations." *arXiv:2404.03002*, 2024.

[2] MICROSCOPE Collaboration. "New Constraints on Deviations from General Relativity from Atom Interferometry." *Physical Review Letters*, 108, 171801, 2012.

[3] Khoury, A., Hinterbichler, W., and Dvaluation, D. "Anchoring the Scalar Field Potential in Chameleon Gravity Models." *Physical Review D*, 68, 022001, 2003.

- [4] Lelli, F., McGaugh, S., and Schombert, J. "SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves." *Astronomical Journal*, 152, 157, 2016.
- [5] Bekenstein, J. D. "Universal Upper Bound on the Entropy-to-Energy Ratio for Bounded Systems." *Physical Review D*, 23, 2873, 1981.
- [6] Boudaud, M. and Cirelli, M. "Voyager 1 further constrain primordial black holes as dark matter." *Physical Review Letters*, 122, 041104, 2019.
- [7] Su, B.-Y. "Constraining primordial black holes as dark matter using AMS-02 data." *European Physical Journal C*, 2024.
- [8] "Updated constraints on primordial black hole evaporation." *Journal of Physics: Conference Series*, 2023.

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